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SYMBIOSIS INSTITUTE OF TECHNOLOGY, PUNE

Symbiosis International (Deemed University)

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Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

Assignment No. 10

Subject:	DevOps Lab
TCODE:	TE7950
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Branch	CSE, Batch (2023-27), L3
Academic Year & Semester	25 – 26, Sem VII
Date of Performance	23rd July, 2026
Title of Assignment:	Docker Volumes & Data Persistence:
Theory:	Docker Volumes provide persistent storage that exists independently of containers. Data stored in a volume remains intact even if a container is stopped, removed, or recreated. Volumes enable applications to retain important data, making them suitable for databases, logs, configuration files, and other persistent storage requirements.
Implementation:	Step 1: Modify the Flask application to store greeting logs inside the `/app/data` directory.

app.py > ...

```python
from flask import Flask
import os
from datetime import datetime

app = Flask(__name__)

DATA_DIR = "/app/data"
PERSISTENCE_FILE = os.path.join(DATA_DIR, "greeting_log.txt")

if not os.path.exists(DATA_DIR):
    os.makedirs(DATA_DIR)

@app.route("/")
def hello_world():
    message = "Hello, Docker!"
    timestamp = datetime.now().strftime("%Y-%m-%d %H:%M:%S")

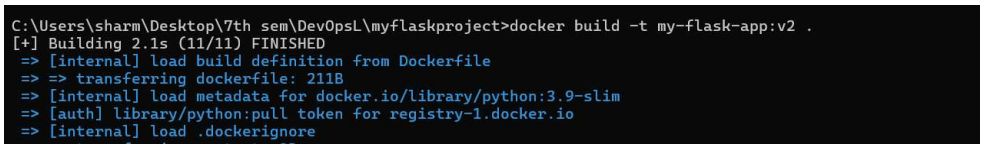
    with open(PERSISTENCE_FILE, "a") as f:
        f.write(f"{timestamp}: {message}\n")

    try:
        with open(PERSISTENCE_FILE, "r") as f:
            log = f.read()
    except FileNotFoundError:
        log = "No previous greetings."

    return f"{message}<br><br><pre>{log}</pre>"

if __name__ == "__main__":
    app.run(host="0.0.0.0", port=5001, debug=True)
```

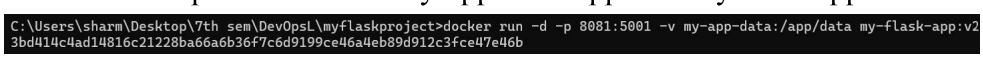
Step 2: Update the Dockerfile (if required) and rebuild the Docker image:
docker build -t my-flask-app:v2 .



Step 3: Create a Docker volume:
docker volume create my-app-data



Step 4: Run the container by mounting the volume:
docker run -d -p 8080:5001 -v my-app-data:/app/data my-flask-app:v2



Step 5: Access the application through the browser and refresh multiple times to generate log entries.



Hello, Docker!

```
2026-07-23 04:37:13: Hello, Docker!
2026-07-23 04:37:14: Hello, Docker!
2026-07-23 04:37:14: Hello, Docker!
2026-07-23 04:37:14: Hello, Docker!
2026-07-23 04:37:15: Hello, Docker!
2026-07-23 04:37:15: Hello, Docker!
2026-07-23 04:37:15: Hello, Docker!
```

Step 6: Stop and remove the container.

```
C:\Users\sharm\Desktop\7th sem\DevOpsL\myflaskproject>docker stop 3bd414c4ad14
3bd414c4ad14

C:\Users\sharm\Desktop\7th sem\DevOpsL\myflaskproject>docker rm 3bd414c4ad14
3bd414c4ad14

C:\Users\sharm\Desktop\7th sem\DevOpsL\myflaskproject>docke ps
'docke' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\sharm\Desktop\7th sem\DevOpsL\myflaskproject>docker ps -a
CONTAINER ID   IMAGE                     COMMAND                  CREATED              STATUS                          PORTS
          NAMES
8d19f1e063c5   my-flask-app:v2           "python app.py"          About a minute ago   Created
          gallant_bose
c646289e3cbc   myflaskapp                "python app.py"          13 minutes ago       Up 13 minutes                   0.0.0.0:8080->5000/tcp, [::]:8080->50
00/tcp    zen_shtern
f4ec9dd44cb0   myflaskapp                "python app.py"          13 minutes ago       Created
          youthful_goodall
f8cdeed68200   myflaskapp                "python app.py"          19 minutes ago       Exited (0) 18 minutes ago
          romantic_mendeleev
```

Step 7: Run another container using the same Docker volume.

```
C:\Users\sharm\Desktop\7th sem\DevOpsL\myflaskproject>docker run -d -p 8080:5000 -v my-app-data:/app/data my-flask-app:v2
54a0135a8baed0247b77df8d15ab22ab0b1ebcbae37cfd6065f81070a54e8dc
```

Step 8: Verify that the previous log entries are still available, demonstrating data persistence.



Hello, Docker!

```
2026-07-23 04:41:18: Hello, Docker!
2026-07-23 04:41:19: Hello, Docker!
2026-07-23 04:41:20: Hello, Docker!
2026-07-23 04:41:20: Hello, Docker!
```